



Evaluating Multimodal Multiplayer Games for Cognitive and Social Skills Development in Dyslexic Children

Project ID: 25-26j-421

Project Proposal

BSc. Special (Hons) in Information Technology, specialized in Interactive Media
Department of Information Technology
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DECLARATION

We declare that this is our own work. This proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning, and to the best of our knowledge and belief, it does not contain any material previously published or written by another person, except where the acknowledgement is made in the text.

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ABSTRACT

Children with dyslexia face persistent challenges in letter recognition and decoding that extend beyond cognitive difficulties to include diminished self-confidence, heightened anxiety, and fear of failure. Traditional literacy interventions and existing digital learning tools predominantly adopt single-player formats that reinforce isolation while failing to address critical psychosocial dimensions necessary for sustained engagement. This research proposes the design, development, and evaluation of a multimodal multiplayer literacy learning system integrating a mobile-based letter interaction application with a PC-based virtual learning environment for children with dyslexia aged 8-12 years.

The system combines real-time multiplayer gameplay, voice-enabled peer communication, anticipatory feedback mechanisms, and synchronized dual-device interactions to support literacy skill development and psychosocial well-being. The mobile companion application (Flutter) enables structured letter tracing and selection activities triggered by in-game events, while the PC-based environment (Unreal Engine) facilitates collaborative word formation tasks. Cloud synchronization via Firebase ensures seamless device communication, while Steam voice integration enables natural peer interaction without visual exposure.

Novel design features include anticipatory feedback providing real-time guidance before errors occur, non-competitive reward mechanisms emphasizing collective achievement, and multimodal learning pathways distributing cognitive load across visual, auditory, and kinesthetic modalities. Unlike existing interventions focusing primarily on decoding accuracy, this research prioritizes confidence, social participation, motivation, and fear of failure alongside literacy performance metrics.

Expected outcomes include empirical evidence on multiplayer collaborative environments for dyslexic learners, design guidelines for dyslexia-sensitive educational games, and insights into how voice-enabled social interaction influences confidence in low-pressure virtual contexts. This research contributes foundational evidence on integrated systems addressing both cognitive processing constraints and emotional-social needs, informing evidence-based design principles for next-generation assistive educational technologies.

Keywords: Dyslexia, Game-Based Learning, Multimodal Interaction, Multiplayer Learning, Educational Technology, Anticipatory Feedback, Assistive Technology

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1. Introduction

Dyslexia constitutes a neurodevelopmental learning difficulty that impairs the acquisition and use of linguistic skills, particularly affecting accurate and fluent word recognition, decoding, and spelling. This condition impacts approximately 5–10% of children worldwide, manifesting in persistent challenges with letter-sound correspondence and phonological processing [1]. Beyond these core academic deficits, dyslexia leads to broader consequences, including reduced self-confidence, heightened fear of failure in literacy tasks, social withdrawal from group learning activities, and diminished motivation for sustained engagement. These emotional and social dimensions exacerbate the condition's effects, as affected children often internalize struggles as personal inadequacies, hindering overall educational progress. Consequently, dyslexia extends far beyond isolated cognitive impairments to influence holistic development in school environments.

Traditional literacy instruction, characterized by repetitive phonics drills and isolated word exercises, proves inadequate for dyslexic children due to its emphasis on linear, text-dominant practice that overlooks variable processing speeds and contextual needs. Such methods frequently induce frustration without fostering transferable skills, as they demand prolonged focus on phonological weaknesses without supportive scaffolds. Transitioning to digital learning tools reveals similar constraints, particularly in single-player formats that prioritize decoding accuracy metrics while neglecting social interaction. These tools reinforce isolation by presenting solitary challenges, failing to counteract motivational decline or emotional barriers associated with repeated errors. Ultimately, both traditional and conventional digital approaches perpetuate cycles of disengagement, underscoring the need for more adaptive, inclusive alternatives [2].

Digital game-based learning (DGBL) offers a promising alternative by integrating multimodal interactions, encompassing visual, auditory, and kinesthetic elements, to support engagement and skill development for dyslexic learners. Through rewards, progression systems, and narrative structures, DGBL cultivates intrinsic motivation, encouraging voluntary practice in low-pressure environments [1]. Evidence from serious games demonstrates improved rhythmic synchronization and attention sustainment, facilitating repeated exposure to literacy-related tasks without overt failure stigma. Immediate feedback loops in these systems normalize trial-and-error processes, transforming potential setbacks into constructive learning opportunities. This multimodal framework thus aligns with dyslexic children's strengths in non-verbal processing, promoting deeper retention than unidirectional methods.

Emerging evidence suggests that incorporating social and multiplayer dimensions into DGBL may further enhance outcomes for dyslexic children, though findings remain exploratory. Initial studies indicate potential benefits from collaborative gameplay, such as peer-supported problem-solving that normalizes mistakes and fosters social confidence [2]. Research hints that group dynamics in game contexts could amplify motivation through shared achievements, yet applications for dyslexia have focused primarily on individual screening rather than interactive literacy practice [3]. These preliminary insights underscore peer collaboration's role in reducing isolation, but conclusive evidence from controlled multiplayer trials specific to dyslexic populations is lacking. Overall, this area invites cautious optimism while highlighting the need for targeted investigation.

Despite advances in DGBL, notable research gaps persist, presenting opportunities for systematic inquiry. Controlled studies of multiplayer literacy games explicitly designed for dyslexic children remain scarce, with existing work favoring single-user prototypes over group-based interventions [2]. Research on connected two-device systems, such as those bridging mobile and PC platforms for seamless cross-context collaboration, is virtually absent, limiting explorations of flexible learning ecosystems. Furthermore, studies overemphasize decoding accuracy as the primary outcome, sidelining confidence-building, social participation, and error normalization critical for long-term engagement [1]. Addressing these voids could refine DGBL's application to dyslexia more comprehensively. Such gaps reflect natural evolutions in an emerging field rather than deficiencies.

This project positions itself to investigate these gaps through an exploratory examination of multiplayer multimodal DGBL systems for dyslexic children, maintaining a focus on research evidence rather than clinical application. It aims to assess whether collaborative gameplay across connected devices supports confidence, motivation, and social participation alongside literacy skill practice. Building respectfully on prior multimodal and preliminary social findings, the work avoids unsubstantiated outcome claims [3]. By prioritizing psychosocial dimensions in a controlled design, the study seeks to generate foundational data informing future IT-driven educational tools. Ultimately, this contribution will advance understanding in this nascent domain, guiding evidence-based enhancements to dyslexia support.

2. Background

Dyslexia is a neurodevelopmental learning difficulty that affects reading and writing skills. It mainly involves problems with phonological processing, which is recognizing sounds in words, letter recognition, decoding words, and holding information in working memory. These issues make it hard for children to keep up in digital learning environments, where they must quickly process text and follow fast-paced tasks. Dyslexia also brings emotional challenges like anxiety about making mistakes, fear of failing in front of others, and low self-belief in learning abilities. Digital tools need to address these by reducing stress and building confidence, leading to designs that support both thinking skills and feelings [4].

Multimodal learning uses more than one sense at once, like sight, sound, and touch. For example, a lesson might show pictures, play audio instructions, and let kids use gestures on a screen. This approach draws on cognitive load theory, which says the brain has limited attention, where too much information at once causes overload, but spreading it across senses makes learning easier. Dual coding theory adds that combining words with images creates stronger memories because the brain processes them in different ways. Children with dyslexia benefit here since their sound-based reading struggles can be helped by strong visual and movement cues. This sets the stage for game-based systems that mix senses smoothly [5].

Digital game-based learning, or DGBL, uses games designed for education, often called serious games. Flow theory explains its power: when a game's challenge matches a player's skill, they enter a focused, enjoyable state without getting bored or frustrated. Games provide quick feedback, like points or hints right after a try, and let players practice safely with retries and no real-world penalties. Studies show DGBL helps children with disabilities stay engaged longer and improve skills through rewards, stories, and step-by-step challenges. For dyslexia, this means turning hard reading tasks into fun levels that keep kids coming back. These ideas naturally lead to adding social elements [1] [A Conceptual Review of Gamification and Game-Based Learning: Synthesizing Evidence Through Self-Determination Theory].

Social constructivist theory says people learn best by working together, building knowledge through talk and shared activities. Vygotsky's zone of proximal development, or ZPD, describes tasks a child can do with help from peers or teachers but not alone. In digital games, peer scaffolding happens when one child hints to another during a joint puzzle, like sharing strategies in a team word challenge. Early studies suggest multiplayer games reduce loneliness for struggling learners by letting kids share mistakes and celebrate wins together. For dyslexia, this could build teamwork skills, though research is still developing. Social features connect to handling errors well [6][7].

Errors play a key role in learning, as they show what to fix next. A growth mindset views mistakes as chances to improve, not signs of failure. Psychological safety means kids feel safe to try without shame, which games support with gentle hints and "keep going" messages. Children with dyslexia often fear errors most because past struggles make them expect to fail. Digital designs can help by offering unlimited practice in private or group settings, slowly building success. This focus on feelings ties into stronger overall learning designs [3][5].

These theories fit together to guide digital learning for dyslexia. Multimodal methods ease thinking demands, games keep motivation high with safe practice, social play adds peer support, and error-friendly designs build confidence. Yet key areas need more study, like combining multiplayer games with multiple senses for dyslexic kids or linking devices for home-school play. These foundations show why new research matters to fill these spaces [2][6].

3. Literature Review

3.1 Dyslexia and Learning Challenges in Digital Environments

Dyslexia, defined as a specific neurodevelopmental learning difficulty, manifests primarily through persistent deficits in accurate and fluent word recognition stemming from phonological processing impairments. Children with dyslexia struggle with mapping sounds to letters, leading to challenges in letter decoding and recognition that hinder early reading development despite adequate intelligence and instruction. These core linguistic difficulties extend to broader cognitive demands, where rapid symbol-to-sound conversion falters under time pressure, a common feature in interactive digital interfaces [8].

Working memory limitations further compound these issues, as dyslexic children exhibit reduced capacity to temporarily hold and manipulate phonological information during tasks requiring sequential processing. Attention and executive function deficits, including poor cognitive flexibility and inhibitory control, exacerbate this by impairing sustained focus amid distractions inherent in screen-based activities. In digital learning environments, these cognitive load challenges become pronounced; for instance, multitasking between visual text, audio cues, and navigation menus overwhelms limited resources, resulting in fragmented comprehension and slower skill acquisition [9] [10].

Emotional and affective factors intensify these cognitive barriers, with dyslexic children often experiencing low reading self-concept, heightened trait anxiety, and pervasive fear of failure linked to repeated decoding frustrations. Such negative self-perceptions create a cycle of avoidance, where anticipated struggles diminish motivation and engagement before tasks begin. Research highlights that trait emotional intelligence may buffer anxiety, yet dyslexic learners frequently lack protective self-belief, leading to withdrawal in evaluative settings [11] (Reading self-concept, trait emotional intelligence and anxiety in students with dyslexia: are reading self-concept and trait emotional intelligence protective factors?).

In digital and game-based learning contexts, these intertwined challenges significantly impair outcomes. Online platforms amplify phonological and working memory demands through scrolling text, timed quizzes, and multimedia integration, fostering disengagement as errors accumulate without supportive scaffolding. Systematic reviews confirm dyslexia reduces effectiveness in virtual environments, where low confidence manifests as hesitancy to interact, perpetuating skill gaps; for example, decoding bottlenecks in game narratives disrupt flow, heightening anxiety and dropout rates. Even collaborative digital games reveal engagement hurdles, as social exposure to errors triggers fear of judgment, underscoring the need for adaptive designs that normalize mistakes [12] [13].

Collectively, these cognitive, linguistic, and emotional dimensions reveal dyslexia's multifaceted impact on digital learning, where traditional interfaces inadvertently exacerbate vulnerabilities rather than mitigate them. Addressing this requires targeted interventions attuned to both processing constraints and psychosocial needs, informing the theoretical foundations explored subsequently [8].

3.2 Multimodal and Game-Based Learning for Dyslexia

Digital interventions for dyslexia have evolved from basic phonics software to sophisticated game-based applications, demonstrating measurable improvements in core reading skills among at-risk children. Early digital tools focused on structured phonics instruction delivered through adaptive interfaces, enabling repeated practice of letter-sound mappings in controlled sessions. A key example involves GraphoGame, a digital phonics program that provided brief daily training to pre-readers identified as cognitively at risk for dyslexia, resulting in significant gains in print knowledge compared to untrained controls [14]. These outcomes highlight how interactive digital formats support phonological awareness by offering immediate feedback and personalization, fostering skill development without the rigidity of traditional worksheets.

Building on these foundations, serious games have emerged as effective tools for targeting specific cognitive deficits associated with dyslexia, such as rhythmic timing linked to reading fluency. One feasibility study tested a rhythm-training serious game with children diagnosed with dyslexia, revealing high usability scores and preliminary evidence of improved synchronization abilities after short-term use [1]. Participants reported enjoyment and ease of use, indicating strong engagement and motivation driven by game mechanics like progressive challenges and visual rewards. These findings affirm that serious games enhance cognitive support through playful repetition, sustaining attention in ways that conventional drills often fail to achieve.

Modern interactive systems further refine this approach by integrating language and rhythm training within accessible platforms, yielding concrete learning outcomes for children with dyslexia-specific reading disorders. A recent evaluation of Mila-Learn, a serious game combining linguistic exercises with rhythmic elements, documented statistically significant improvements in reading accuracy and fluency following consistent play sessions [15]. The game's design emphasized short, engaging modules that boosted motivation, as evidenced by high completion rates and positive qualitative feedback from users. Such results underscore the potential of multimodal digital games to address interconnected dyslexia symptoms, promoting both cognitive gains and voluntary participation.

Mobile educational games represent another advancement, leveraging portability and structured literacy methods to deliver interventions remotely. An educational mobile application game based on Orton-Gillingham principles, emphasizing multisensory phonics, was assessed with dyslexic children, showing enhanced reading proficiency and sustained engagement through telepractice delivery.[16]. Quantitative measures indicated progress in word recognition alongside qualitative reports of increased confidence, attributable to the game's intuitive controls and achievement-based progression. This approach demonstrates usability in real-world settings, extending dyslexia support beyond clinical environments.

Despite these strengths, existing game-based interventions exhibit notable limitations that temper their broader impact. Most studies report primarily single-player experiences, which effectively build isolated skills like phonics or rhythm but overlook opportunities for social interaction that could normalize errors and enhance peer learning [14][1]. Engagement relies heavily on intrinsic motivators such as rewards and adaptation, yet lacks immersive multimodal depth, such as integrated gesture or collaborative features, limiting transfer to complex literacy tasks. Moreover, while usability and short-

term outcomes are promising, few evaluations occur in fully controlled, longitudinal environments, leaving questions about sustained efficacy unanswered [15][16].

These interventions thus provide a solid empirical base, improving phonological skills, motivation, and usability, but reveal gaps in social integration and comprehensive evaluation. Addressing such shortcomings points naturally to the value of multiplayer and collaborative approaches in subsequent sections.

3.3 Social and Collaborative Learning in Multiplayer Digital Games

Social constructivist learning theory provides a strong theoretical foundation for multiplayer game-based learning, positing that knowledge emerges through dialogue and collaborative activity rather than isolated effort. Vygotsky's zone of proximal development (ZPD) describes tasks children can accomplish with peer guidance but not independently, while peer scaffolding in digital environments involves one learner offering hints or strategies during joint challenges. Self-determination theory complements this through its "relatedness" need, where social connections drive intrinsic motivation. For children with dyslexia, who often face isolation from literacy struggles and fear of judgment in group settings, these frameworks suggest multiplayer interactions could create safe practice spaces that normalize difficulties and build confidence. Theoretical models thus indicate that collaborative digital play may counteract affective barriers like anxiety, fostering engagement through shared success. While these frameworks strongly suggest multiplayer interaction could benefit dyslexic learners, direct empirical evidence remains limited, with most existing research examining general learning disabilities or special education contexts [2].

Research on multiplayer game-based learning in related populations provides suggestive evidence for potential benefits transferable to dyslexia. Studies of learners with disabilities show collaborative digital games improve engagement and social participation by enabling peer idea exchange during shared tasks, reducing isolation compared to solo formats [12]. A metasynthesis of game applications for dyslexic reading proficiency notes collaborative elements promote motivation through interaction, though multiplayer specifics are not central [17]. Findings from special education contexts indicate gamified group dynamics enhance skill development via mutual support, suggesting potential for dyslexic children facing similar motivational declines. While these studies do not focus exclusively on dyslexia, they suggest multiplayer dynamics may address parallel affective challenges, including isolation, fear of failure, and reduced persistence, common among dyslexic learners [18].

Multiplayer mechanisms could particularly aid dyslexic children by redistributing cognitive and emotional demands during literacy practice. Error normalization occurs when cooperative play frames mistakes as group learning opportunities. For instance, peers collectively debug a word puzzle rather than one child failing alone, directly countering dyslexia's heightened fear of decoding errors. Distributed cognition lightens individual working memory loads, as one child identifies letter patterns while another handles pronunciation, easing phonological bottlenecks. Social accountability from peer presence sustains motivation through shared goals, like advancing team levels, while collaborative success shifts evaluative pressure from personal performance to collective

achievement, potentially lowering anxiety. For dyslexic learners, who experience repeated frustrations in isolated settings, these dynamics may provide psychosocial safety absent in single-player tools, encouraging risk-taking in reading tasks [17][18].

Existing research reveals critical gaps that necessitate further investigation into multiplayer designs for dyslexia. Most studies employ exploratory or qualitative methods rather than controlled experiments, limiting causal evidence on learning outcomes [2]. Real-time synchronous multiplayer environments tailored to dyslexic cognitive profiles remain underexplored, with social features rarely integrated into targeted phonological or decoding activities [17]. Long-term literacy transfer and sustained confidence gains lack rigorous evaluation, particularly in special education applications [18]. Multimodal multiplayer systems combining social dynamics with evidence-based literacy instruction are absent, as are designs addressing dyslexia's unique affective needs like error-safe collaboration. These gaps reveal a clear opportunity: controlled evaluation of multiplayer, multimodal literacy games designed specifically for children with dyslexia, integrating social interaction with evidence-based literacy instruction. This foundation establishes the rationale for examining such integrated systems.

3.4 Motivation, Feedback, and Emotional Regulation in Learning Games

Learning games influence motivation and engagement primarily through carefully designed feedback and reward systems that transform challenging cognitive tasks into rewarding experiences. Real-time feedback, delivered immediately after actions, guides learners toward correct responses while maintaining flow, whereas post-error feedback helps reframe mistakes constructively to prevent disengagement. For neurodiverse children, including those with dyslexia, these mechanisms prove essential, as games with progressive difficulty and visual rewards sustain attention during repetitive phonological practice that might otherwise provoke frustration [1]. Reward structures, such as achievement badges or level unlocks, tap into intrinsic motivation by signaling competence, encouraging persistence even when initial attempts fail. This interplay fosters voluntary practice, particularly valuable for dyslexic learners who face motivational barriers from repeated decoding struggles.

Feedback strategies vary in their impact on persistence and emotional responses, with studies highlighting the need for balanced approaches tailored to cognitive vulnerabilities. Anticipatory feedback, which previews challenges or offers hints before errors occur, remains underexplored but shows promise in reducing initial frustration by building proactive confidence. Real-time cues, like rhythmic synchronization prompts in serious games, maintain engagement by aligning immediate actions with learning goals, leading to high usability ratings among dyslexic children during home-based sessions [1]. Post-error feedback dominates current designs, providing explanatory hints after mistakes to reinforce learning paths, yet it risks reinforcing failure awareness if not paired with positive reinforcement. Systematic reviews confirm that such feedback enhances persistence for learners with disabilities, as adaptive responses prevent overload and promote iterative improvement [2].

Emotional regulation emerges as a key benefit of well-designed learning games, directly addressing

fear of failure and frustration common in neurodiverse populations. Games create low-stakes environments where errors trigger supportive messages rather than penalties, normalizing setbacks and gradually building resilience, which is crucial for dyslexic children prone to anxiety from phonological deficits[19]. Positive reinforcement through encouragement and progress visualization supports confidence by emphasizing growth over perfection, with empirical evidence showing sustained engagement in math games for learning-disabled students via achievement feedback [20]. These elements manage frustration by distributing emotional load across playful mechanics, fostering self-regulation as learners experience mastery cycles. For dyslexia specifically, accessible designs with customizable feedback reduce dropout risks, though studies note inconsistent measurement of long-term emotional outcomes.

Despite these advances, significant limitations persist in the literature, underscoring gaps that warrant further exploration. Research over-relies on post-error feedback, neglecting anticipatory systems that could preempt frustration in real-time dyslexia interventions [12]. Controlled experimental comparisons between feedback types remain scarce, with most evaluations relying on usability surveys rather than longitudinal emotional metrics [1]. Emotional outcomes beyond surface-level engagement, such as sustained confidence or reduced test anxiety, receive insufficient attention, particularly for dyslexic populations where affective barriers hinder skill transfer [19]. Neurodiverse studies often generalize findings without dyslexia-specific controls, leaving uncertainty about reward mechanisms' efficacy for phonological persistence [20].

These insights from existing research thus inform the design of future learning games by highlighting the need for integrated, emotionally attuned feedback that normalizes mistakes through anticipatory guidance and adaptive reinforcement. This literature establishes a clear rationale for developing systems that prioritize both cognitive support and affective resilience in dyslexia interventions.

4. Research Gap

4.1 Lack of Integrated Multi-Device Learning Environments

Digital game-based learning has gained traction as a promising intervention for children with dyslexia and neurodiverse populations, leveraging multimodal interactions to enhance engagement and address reading challenges. Existing research highlights the potential of interactive games, including those incorporating visualization and virtual reality, to foster motivation and social skills in neurodevelopmental contexts akin to dyslexia [21][22]. Studies often emphasize qualitative benefits such as improved social sharing through turn-taking mechanics or cooperative play, drawing from small-scale designs targeted at autistic children whose needs overlap with dyslexic learners in terms of attention and interaction deficits [23] [24]. These works underscore how game elements like shared virtual spaces can promote peer collaboration, yet they predominantly rely on usability testing, expert feedback, or prototype evaluations rather than rigorous quantitative analysis.

While some investigations introduce collaborative features, such as turn-based multiplayer formats, the field remains anchored in single-player or guided individual experiences, with multiplayer dynamics underexplored under controlled conditions [21][23]. For instance, literature reviews reveal enthusiasm for VR-supported serious games that enable communication in multiplayer settings for special needs children, but these syntheses expose a pattern of short-term observations without experimental validation comparing collaborative to isolated play [22]. Similarly, cooperative vocabulary games prioritize design for social vocabulary building yet stop at preliminary evaluations, neglecting structured assessments of how peer interactions influence learning persistence or emotional outcomes [24]. This reliance on qualitative insights from prototypes limits generalizability, as interventions seldom scale to controlled trials measuring cognitive gains alongside motivational shifts.

Critically, direct empirical comparisons between multiplayer and single-player modalities are scarce, leaving unanswered whether social elements amplify engagement or mitigate isolation common in dyslexic learning. Surveys of interaction technologies note multimodal games' appeal for neurodiverse engagement but fail to quantify peer collaboration's impact relative to solitary modes, highlighting methodological gaps in outcome measurement [21]. Turn-taking games demonstrate feasibility for social skill-building, yet without baseline single-player controls or longitudinal tracking of confidence and motivation, their superiority remains speculative [23]. Comprehensive reviews further confirm that while multiplayer VR holds promise for shared learning, the absence of quantitative data on social-motivational effects perpetuates a disconnect between design innovation and evidential support [22].

These limitations collectively point to a pressing research gap: the need for controlled experimental evaluations of multiplayer learning environments tailored to dyslexic children, incorporating quantitative metrics for social interactions, motivational outcomes, and comparative efficacy against single-player approaches. Addressing this void through structured investigations would bridge the divide between promising collaborative designs and robust evidence, informing more effective interventions that harness peer dynamics to overcome dyslexia-related barriers [23][24].

4.2 Underexplored Multiplayer and Social Learning for Dyslexia

Digital games for dyslexia and neurodiverse learners increasingly incorporate multimodal elements, such as handwriting, tracing, and visual-motor tasks, to address cognitive and perceptual challenges in reading and writing. Research demonstrates potential in activities like spelling assessments comparing handwriting to typing modes, revealing mode-specific effects on accuracy for dyslexic children, yet these remain confined to isolated, non-game contexts [25]. Similarly, HCI studies critique single-modality designs in games for neurodivergent populations, including those with dyslexia-related dysgraphia, where drawing or motor inputs are underexplored beyond solo tablet interactions [26]. Participatory design efforts with neurodiverse children introduce simple multimodal kits, like robot-based games, but limit evaluation to small-group, single-platform sessions without deeper cognitive load analysis [27].

Existing approaches predominantly operate within single-device ecosystems, evaluating multimodal tasks like handwriting legibility or eye-tracking puzzles in solo settings that fail to leverage cross-device synergies. For instance, studies on dyslexic handwriting difficulties highlight motor slowness and unsteadiness in dictation tasks but isolate these from dynamic game environments or multi-device inputs that could distribute cognitive demands [28]. Reviews of serious games for Arabic dyslexia employ multimodal tracking in puzzle formats on one device, noting engagement benefits yet overlooking how mobile inputs might influence shared PC-based visuals or collaborative tracing activities [29]. Critically, HCI critiques reveal that neurodivergent game research favors individual device use, neglecting real-time integration of motor-visual tasks across platforms and absent rigorous cognitive load metrics during such interactions [30].

This pattern underscores underexplored synergies, such as the combined impact of drawing-based motor inputs with visual recognition in multi-device setups, where a tablet's tracing could dynamically alter a shared screen's feedback loop. Few studies embed these modally rich tasks into multiplayer contexts, leaving questions about engagement boosts or load reductions from peer-synchronized, cross-device play unaddressed [27][29]. Moreover, controlled comparisons pitting multimodal, multi-device systems against standard single-player games are scarce, limiting evidence on whether distributed interactions mitigate dyslexia-specific overloads better than isolated modalities [25][28].

Addressing these gaps through targeted investigation of multimodal, multi-device learning environments is essential to advance dyslexia interventions. Such research would clarify how integrated interactions enhance cognitive processing and sustained engagement, providing empirical foundations for scalable, inclusive games that transcend single-platform limitations [26][30].

4.3 Overreliance on Post-Error Feedback Mechanisms

Existing research on feedback in digital learning games for neurodiverse and dyslexic children predominantly employs post-error corrective mechanisms, delivering guidance only after mistakes

occur to reinforce learning through immediate or delayed responses [31][32]. These studies emphasize static or rule-based feedback triggered reactively, such as low-latency corrections following task completion, which aids retention in game-like settings but overlooks predictive interventions that anticipate errors during execution [33][34]. Multimodal or adaptive elements remain underexplored, with emotional dimensions like frustration, anxiety, or confidence rarely quantified, limiting insights into how real-time scaffolding could mitigate fear of failure in vulnerable learners [31][32].

This reliance on retrospective feedback reveals a critical gap: no reviewed study integrates anticipatory, real-time mechanisms, such as error prediction via multimodal cues, within multiplayer or socially interactive games tailored for dyslexic children[33][34]. Addressing this void is essential, as anticipatory guidance in social contexts could enhance learning effectiveness by preempting errors, foster emotional regulation to reduce anxiety, and sustain motivation through confidence-building peer interactions, ultimately yielding more robust interventions for dyslexia [31] [32].

4.4 Limited Focus on Emotional Regulation and Fear of Failure

Research on digital learning environments for children with dyslexia has explored multiplayer and collaborative elements primarily through tutor-guided or small-group play, revealing spontaneous social interactions like "game talk" that enhance engagement during literacy tasks [12]. However, these studies remain largely single-player or non-peer-based, with limited empirical validation of real-time multiplayer dynamics alongside voice communication or dual-device setups [1][35]. Voice or audio-supported interactions are underexplored in dyslexia contexts, while social participation benefits are suggested through gamified reviews, yet communication anxiety and fear of failure receive insufficient integration into controlled, immersive designs [2].

Prior work examines multiplayer collaboration, voice elements, and emotional safety in isolation, with few systems combining these in dyslexia-focused literacy games that leverage cross-platform interactions for reduced social anxiety [12][1]. Emotional confidence outcomes, such as mitigated fear of failure, are rarely measured quantitatively in socially interactive settings, leaving largely unexplored the synergistic potential of voice-enabled multiplayer environments [2][35]. This insufficient integration highlights a critical need for comprehensive studies validating multiplayer, voice-driven, dual-device systems to foster emotional safety and sustained social participation in dyslexia interventions.

4.5 Lack of Integrated Dual-Device and Multiplayer Learning Systems for Dyslexia

Existing research has explored dual-device and companion-app configurations to enhance engagement and usability in educational settings. Studies demonstrate how synchronous mobile collaboration supports group interactions in computer-supported learning environments [Exploring group interactions in synchronous mobile computer-supported collaborative learning with mobile devices]. Mobile educational apps have been found to boost relational competencies and learner engagement when used alongside traditional methods [36]. Second-screen applications improve

attention allocation in multimodal tasks, particularly under medium cognitive loads [37]. Reviews of cross-device media highlight synchronization benefits for companion interactions in ubiquitous computing contexts [38].

While these works address usability in multi-device learning and multimodal engagement, they remain fragmented. None integrate dual-device setups with multiplayer literacy gameplay, nor do they target children with dyslexia [39][36]. Evaluations focus on general attention and competencies rather than social confidence, communication skills, or fear of failure mitigation [37][38].

This leaves critical gaps, including synchronized mobile-PC interactions for literacy tasks, voice-supported cooperative learning in dual-device environments, and controlled assessments of enjoyment from such systems. The proposed research addresses this by developing an integrated multiplayer, voice-enabled dual-device literacy game tailored for dyslexic children.

5. Research Problem

Children with dyslexia face persistent difficulties in letter recognition, decoding, and spelling that extend beyond cognitive challenges to encompass significant emotional and social barriers. Research confirms that dyslexic learners frequently experience diminished self-confidence, heightened anxiety around literacy tasks, and fear of failure that compounds their academic struggles [4][11]. These affective dimensions contribute to reduced social participation in learning environments, creating cycles of isolation and disengagement that traditional interventions fail to address adequately [12][13].

Current digital learning interventions for dyslexia predominantly adopt single-player formats that prioritize isolated skill development, such as phonological awareness or decoding accuracy, while neglecting the social and emotional contexts critical for sustained engagement and confidence-building [2][14]. This emphasis on individual practice inadvertently reinforces isolation, as children encounter literacy challenges alone without the normalizing effect of peer interaction or collaborative problem-solving. Furthermore, existing systems overwhelmingly rely on post-error feedback mechanisms that highlight mistakes after they occur, potentially reinforcing fear of failure rather than preemptively supporting learners through anticipatory guidance [31][32].

A critical gap exists in the integration of multiplayer dynamics with multimodal literacy interventions specifically designed for dyslexic children. While preliminary evidence from related populations suggests that collaborative gameplay may enhance motivation and reduce social anxiety through shared experiences [12][17], controlled investigations of synchronous multiplayer environments tailored to dyslexia-specific cognitive and emotional needs remain virtually absent from the literature [2][18]. The potential benefits of peer scaffolding, error normalization through cooperative play, and distributed cognitive load across collaborative tasks have not been systematically examined in dyslexia contexts, leaving fundamental questions unanswered about whether and how social interaction mechanisms influence literacy engagement and confidence outcomes.

Additionally, current research has not explored the synergistic potential of cross-device multimodal interactions, such as mobile-based tactile literacy activities integrated with immersive shared digital environments, for dyslexic learners. Studies examine mobile educational applications and dual-device configurations in isolation, without investigating how synchronized mobile-to-PC interactions might support collaborative literacy practice while distributing cognitive demands across modalities [36][37][38]. The absence of empirical evidence on voice-enabled communication within multiplayer literacy games represents another unexplored dimension, despite theoretical support for how verbal peer interaction might reduce communication anxiety and foster social confidence [12][35].

These converging gaps reveal a fundamental limitation in current knowledge: we lack empirical understanding of how integrated multiplayer, multimodal learning systems affect not only literacy skill development but also the critical psychosocial dimensions, such as confidence, motivation, fear

of failure, and social participation, that determine long-term engagement for children with dyslexia. Existing interventions target cognitive outcomes without systematically investigating whether collaborative, cross-device, voice-enabled gameplay environments can address the interconnected emotional and social barriers that perpetuate learning difficulties.

This research therefore addresses a distinct knowledge gap by investigating whether and how a multiplayer, voice-enabled, dual-device literacy learning system influences engagement, social confidence, error perception, and motivation in children with dyslexia. Rather than assuming effectiveness, this exploratory study seeks to generate foundational evidence on the mechanisms through which collaborative multimodal gameplay might support both literacy practice and psychosocial well-being. By systematically examining these unexplored dimensions, the research aims to inform evidence-based design principles for next-generation digital learning interventions that holistically address the cognitive, emotional, and social needs of dyslexic learners.

6. Objectives

6.1 Main Objective

The main objective of this research is to design, develop, and evaluate an immersive, multiplayer literacy learning system for children with dyslexia that integrates mobile-based letter interaction with a shared virtual learning environment. The project aims to support literacy development while simultaneously addressing social confidence, motivation, and emotional regulation by providing a safe, collaborative, and engaging gameplay experience. By combining multimodal interaction, cooperative learning, anticipatory feedback, and voice-supported communication, the system seeks to reduce fear of failure and encourage positive learning behaviors among dyslexic children.

6.2 Specific Objectives

1. To design and implement a mobile application that enables children to interact with letters through structured tracing and selection activities, supporting motor memory, personalization, and individual learning preferences.
2. To develop a real-time multiplayer virtual learning environment where children can collaboratively engage in literacy-related tasks, promoting peer interaction, cooperation, and shared problem-solving.
3. To evaluate the impact of a multimodal multiplayer experience on attention, engagement, and learning performance when compared to traditional single-player literacy games.
4. To incorporate an anticipatory feedback mechanism that provides real-time, supportive guidance before errors occur, aiming to reduce frustration and improve task accuracy and learning retention.
5. To integrate a non-competitive reward system and real-time voice communication to enhance motivation, persistence, social confidence, and emotional comfort during learning activities.
6. To assess how the combined system influences children's confidence, social participation, and fear of failure within a virtual learning context.
7. To collect and analyze gameplay and interaction data to understand how multimodal, collaborative learning environments can better support children with dyslexia.

7. Methodology

7.1 Design and Development Approach

7.1.1 Mobile-Based Letter Interaction Design

This subsection describes the development of a mobile application used for letter tracing and selection activities. The application supports multiple difficulty levels, guided tracing, and color-based customization to accommodate individual learning preferences. The mobile device functions as an interactive input medium that responds dynamically to events occurring within the multiplayer game environment.

7.1.2 Multiplayer Learning Environment Design

This subsection outlines the implementation of a real-time multiplayer environment in which children connect and participate in a shared online learning space. Literacy-related tasks are embedded within exploration-based gameplay, requiring players to collect and organize letters collaboratively to form meaningful words. The design emphasizes cooperative interaction and shared task completion.

7.1.3 Anticipatory Feedback Integration

This subsection explains the incorporation of an anticipatory feedback mechanism that provides real-time guidance during letter interaction tasks. Visual, audio, and haptic cues are triggered to gently redirect user actions before incorrect selections occur, supporting smoother task completion and reducing interruptions during gameplay.

7.1.4 Motivation and Communication Support

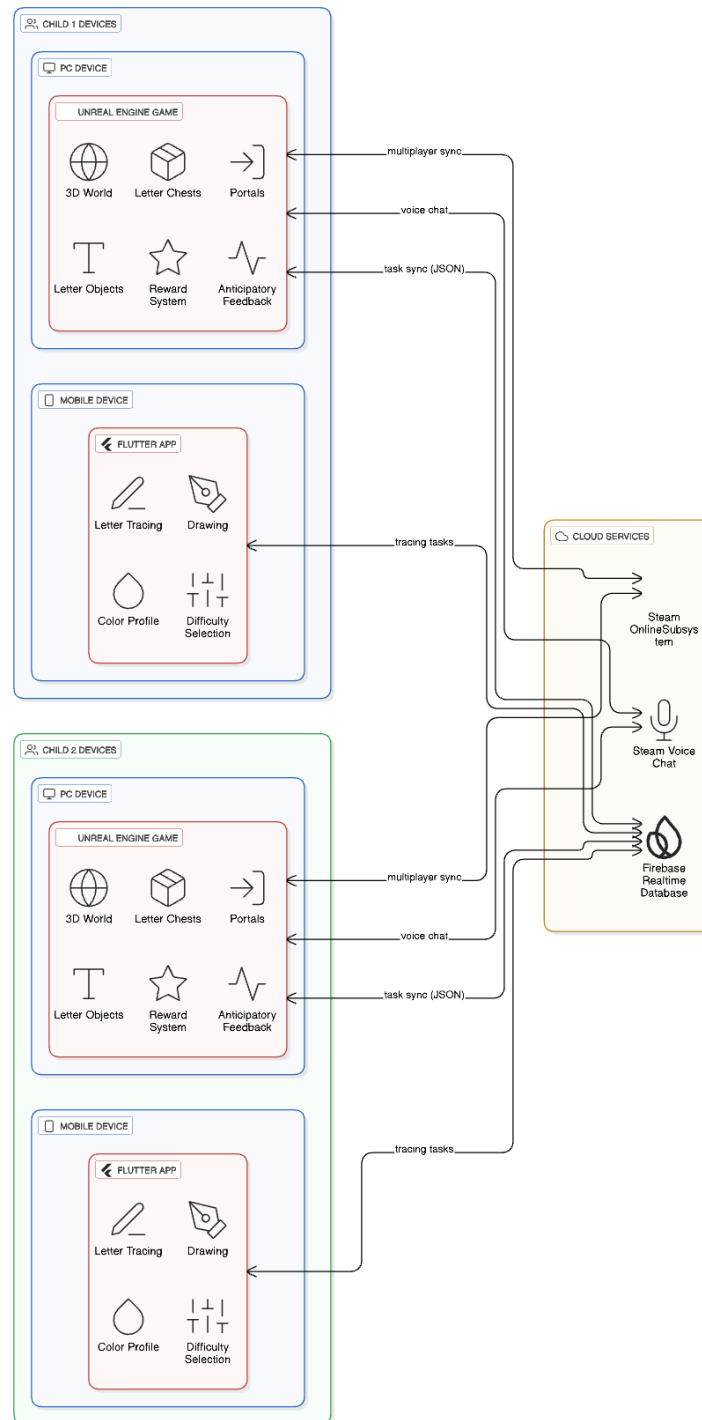
This subsection describes the integration of a reward-based motivation system and in-game voice communication. Rewards are presented as non-competitive visual indicators of progress and effort, while voice communication enables natural interaction between players during collaborative gameplay.

7.2 System Architecture

The proposed system adopts a distributed, multimodal architecture that integrates a PC-based multiplayer game environment with a mobile companion application to support collaborative literacy learning. The architecture is designed to enable real-time interaction between players, synchronized task execution across devices, and seamless communication among participants. Core system responsibilities are divided across the PC game system, mobile application, cloud middleware, and voice communication services to ensure scalability, responsiveness, and reliable synchronization.

The PC game environment acts as the authoritative controller of the shared game world, managing multiplayer sessions, world state, and progression logic. The mobile application operates as an interactive learning interface, delivering letter-based tasks aligned with in-game events. A cloud-based middleware layer facilitates bidirectional communication between the two platforms, while

integrated voice communication supports real-time social interaction among players. This architectural design ensures clear separation of concerns while enabling coordinated, immersive gameplay experiences.



The **PC Game System**, developed using the Unreal Engine, functions as the central multiplayer environment. A designated PC host operates as the listen server, maintaining the authoritative game state, including world progression, chest locations, and letter distribution. Multiplayer session

management and state replication are handled through the Steam Online Subsystem, enabling multiple PC clients to join the same game session and receive synchronized updates.

Game events such as chest discovery or task initiation trigger corresponding learning activities on the mobile companion application. These triggers are transmitted from the PC host to the cloud middleware layer, which relays task requests to the appropriate mobile devices. The system supports session-based linkage between devices through a unique connection code, ensuring correct pairing between players and mobile applications.

The **Mobile Companion Application**, developed using Flutter, is responsible for executing letter-based learning tasks, including tracing, recognition, and difficulty-adjusted activities. Each mobile device listens for task triggers via real-time cloud updates and presents the appropriate activity to the user. Upon successful task completion, the application sends a completion flag back to the cloud middleware, which then notifies the PC game system to update the shared game state accordingly.

The **Cloud Middleware**, implemented using Firebase Realtime Database, serves as the communication bridge between the PC game system and mobile applications. It enables low-latency data exchange using JSON-based event updates and supports real-time synchronization of task triggers and completion states. This layer ensures device independence while maintaining consistent game progression across all participants.

To support social interaction and collaboration, **voice communication** is enabled through the Steam Overlay Voice system. This allows players to communicate in real time within a shared voice channel without visual exposure, reducing social anxiety while promoting cooperation and coordination during gameplay.

Additionally, an **event logging module** within the PC game system records timestamped gameplay and interaction events. These logs support later analysis of player behavior, collaboration patterns, and task completion timelines, contributing to research evaluation and performance assessment.

7.3 Software Solution

7.3.1 PC-Based Multiplayer Game Application

The PC-based game application is developed using the Unreal Engine and serves as the core interactive environment for the learning experience. It manages the shared virtual world, multiplayer sessions, player navigation, game progression, and letter discovery mechanics. One player hosts the session as a listen server, maintaining the authoritative game state while synchronizing updates across all connected clients.

The game logic controls chest placement, portal traversal, guided navigation elements, and letter collection workflows. When players interact with key objects such as chests, the game system triggers corresponding learning tasks on linked mobile devices through cloud-mediated signals. The

Unreal Engine's networking framework, combined with the Steam Online Subsystem, enables reliable session management, state replication, and real-time player synchronization.

An event logging mechanism is embedded within the game to capture timestamped gameplay actions, task triggers, and completion events. These logs support later analysis of gameplay flow, collaboration dynamics, and learning-related interactions for research evaluation purposes.

7.3.2 Mobile Companion Learning Application

The mobile companion application is developed using Flutter to ensure cross-platform compatibility and consistent user experience across Android devices. This application functions as the primary interface for executing letter-based learning activities triggered by in-game events. Upon receiving a task request from the game system, the mobile application presents the appropriate activity based on the selected difficulty level.

The application supports multiple learning modes, including guided letter tracing, reduced-assistance tracing, and recognition-based tasks involving objects or words associated with specific letters. Visual customization options, such as color profiles, are provided to enhance accessibility and personalization for individual learners.

Task progress is monitored locally within the application, and successful completion events are transmitted back to the cloud middleware in real time. This feedback loop allows the PC game system to respond immediately by unlocking chests, updating world progression, and enabling continued gameplay. The mobile application operates independently of the PC platform while remaining synchronized through cloud-based communication.

7.3.3 Cloud Integration and Communication Services

Cloud-based services are employed to facilitate communication and synchronization between the PC game application and mobile companion devices. Firebase Realtime Database is used as the primary middleware layer, enabling low-latency data exchange through event-driven updates. Task triggers, session identifiers, and completion flags are transmitted as structured data to ensure consistent interpretation across platforms.

Device pairing is achieved through a session-based connection code generated by the PC game system and entered into the mobile application at startup. This mechanism ensures correct association between players and their corresponding mobile devices while maintaining simplicity and security.

Voice communication is integrated using the Steam Overlay Voice system, allowing players to communicate verbally during gameplay without visual exposure. This feature supports collaborative interaction in a low-pressure virtual environment, encouraging social participation while reducing fear of failure or peer judgment.

7.3.4 Voice Communication and Reward Mechanisms

The system integrates real-time voice communication and an in-game reward mechanism to support social interaction, motivation, and sustained engagement within the multiplayer learning environment. Voice communication is implemented using the Steam Overlay Voice system, allowing players to communicate verbally during gameplay while remaining visually anonymous. This design choice creates a low-pressure virtual space where children can interact, collaborate, and coordinate without the social anxiety commonly associated with face-to-face communication.

Voice interaction enables spontaneous cooperation, peer support, and shared problem-solving as players navigate the environment, locate letters, and arrange them to form meaningful words. By embedding voice communication within a game-based context, the system encourages natural conversational exchanges that can help reduce fear of failure and support the gradual development of social confidence.

The reward mechanism is designed to reinforce positive engagement rather than competitive comparison. Instead of leaderboards or ranking systems, players receive visual rewards such as stars or completion indicators upon successfully forming words and completing learning tasks. Completion time is presented as reflective feedback rather than as a competitive metric, allowing players to recognize progress without social pressure.

Rewards are triggered by verified task completion events received from the mobile application and are synchronized across all connected clients. This ensures that achievements are shared experiences, reinforcing collaboration and collective success. The combined use of voice communication and non-competitive rewards supports emotional regulation, encourages persistence, and fosters a cooperative learning atmosphere aligned with the needs of children with dyslexia.

7.4 Requirement Analysis, Feasibility, and Data Considerations

This section outlines the process used to identify system requirements, assess feasibility, and determine the data needs required to support the development and evaluation of the proposed multimodal multiplayer learning system for children with dyslexia.

7.4.1 Requirement Gathering and Analysis

Requirement gathering was conducted through literature review, expert consultation, and analysis of existing digital learning tools designed for children with dyslexia.

Educational and Cognitive Requirements:

- Identification of common learning challenges faced by children with dyslexia, including difficulties in letter recognition, sequencing, attention regulation, and reduced confidence in social learning contexts.
- Consideration of emotional and motivational factors such as fear of failure, frustration during literacy tasks, and social withdrawal.
- Emphasis on learning environments that normalize mistakes and encourage collaborative problem solving rather than individual performance pressure.

- Requirement for adaptive difficulty levels in letter tracing and selection tasks to accommodate varying skill levels.
- Integration of multisensory learning principles (visual, auditory, and kinesthetic interaction).

System and Interaction Requirements:

- Support for real-time multiplayer gameplay within a shared virtual environment.
- Seamless interaction between the PC-based game and the mobile companion application using session-based synchronization.
- Reliable triggering of mobile learning tasks from in-game events and validation of task completion before progression.
- Continuous voice communication to support peer interaction and cooperative play.
- Event logging mechanisms to capture gameplay actions, task completion, and interaction patterns for research analysis.

User Requirements:

- Child-friendly interface design suitable for ages 8–12.
- Minimal text dependency to reduce cognitive load.
- Simple onboarding through guided tutorials.
- Clear visual cues and feedback to guide navigation and task completion.
- Safe and supportive social interaction without exposing personal identities.

7.4.2 Feasibility Study

Technical Feasibility:

The research team possesses experience in game development, mobile application development, and cloud-based systems. The use of Unreal Engine for multiplayer game development, Flutter for the mobile application, Firebase Realtime Database for synchronization, and Steam Online Subsystem for networking and voice communication ensures reliance on stable, well-documented technologies. Preliminary prototypes confirm reliable two-device synchronization and real-time task triggering.

Operational Feasibility:

The system is designed to operate in controlled environments such as schools, clinics, or supervised home settings. The gameplay mechanics are intuitive and require minimal technical setup. Hosting and joining sessions are simplified, and mobile device pairing is achieved through a short session code. The system supports small group play, making it suitable for observational research and guided interventions.

Financial Feasibility:

The project is developed using primarily open-source or free-tier technologies. Hardware requirements are limited to standard Windows PCs, Android mobile devices, and basic audio peripherals. Cloud infrastructure costs are minimized through Firebase's free tier during development and testing. As an academic research project, overall development costs remain within reasonable limits.

Ethical and Research Feasibility:

The system does not collect personally identifiable information. Gameplay data focuses on task performance, interaction timing, and engagement metrics. Participation can be conducted with parental consent and child assent, ensuring ethical compliance for studies involving children.

7.4.3 Dataset Requirements

Unlike data-intensive machine learning systems, this project relies primarily on **interaction and performance data** generated during gameplay for evaluation purposes.

Gameplay and Interaction Data:

- Task completion status (success/failure)
- Time taken to complete tracing and selection tasks
- Number of attempts per task
- Sequence of letter collection and word formation
- Session duration and cooperative task completion events

Multiplayer Interaction Data:

- Number of connected players per session
- Frequency and duration of cooperative actions
- Timing of voice communication relative to task events
- Synchronization events between PC and mobile devices

Engagement and Behavioral Metrics:

- Voluntary time spent in gameplay sessions
- Reattempt rates after incorrect actions
- Completion rates across difficulty levels
- Response patterns to reward feedback

Data Handling Considerations:

- All data is anonymized and session-based.
- Data is stored securely using cloud infrastructure.
- Logged data is exported for statistical and qualitative analysis.
- No audio recordings are stored; only interaction metadata is captured.

These datasets enable evaluation of learning performance, engagement, collaboration, and emotional response patterns without requiring large-scale external datasets.

8. Project Requirements

This section outlines the functional, non-functional, user, software, and hardware requirements necessary to design, develop, and evaluate the proposed multimodal multiplayer learning system for children with dyslexia.

8.1 Functional Requirements

The system shall provide the following core functionalities:

- The system shall allow one player to host an online game session and enable other players to join the same environment.
- The system shall support real-time multiplayer interaction, ensuring all connected players share a synchronized world state.
- The system shall enable players to explore the environment, locate letter-containing chests, and collect letters collaboratively.
- The system shall support letter arrangement mechanics, allowing players to form meaningful words using collected letters.
- The system shall provide a mobile companion application that connects to the PC game using a secure session code.
- The mobile application shall present letter tracing and selection tasks based on the signals received from the PC game.
- The system shall validate task completion from the mobile application before unlocking corresponding in-game content.
- The system shall provide real-time voice communication between players during gameplay.
- The system shall provide a reward mechanism that visually acknowledges successful task completion.
- The system shall log gameplay events, task completion times, and player interactions for research analysis.

8.2 Non-Functional Requirements

The system shall satisfy the following quality attributes:

- The system shall maintain low latency to ensure smooth real-time multiplayer interaction.
- The system shall ensure data consistency across all connected clients.
- The system shall be scalable to support multiple players in a single session.
- The system shall be reliable, preventing loss of session data during gameplay.
- The system shall ensure secure communication between the PC game, mobile application, and cloud middleware.
- The system shall be usable by children aged 8-12, with intuitive controls and clear visual feedback.
- The system shall provide fault tolerance to handle temporary network interruptions.
- The system shall support cross-device synchronization without requiring advanced technical setup by users.

8.3 User Requirements

The system is designed to meet the needs of the following user groups:

Primary Users (Children with Dyslexia):

- The system shall provide an engaging, low-pressure learning environment.
- The system shall allow children to collaborate and communicate without fear of public failure.
- The system shall provide supportive feedback rather than punitive error messages.
- The system shall support different difficulty levels to accommodate individual learning needs.

Secondary Users (Researchers / Facilitators):

- The system shall allow monitoring of gameplay sessions for research purposes.
- The system shall provide access to logged performance data for analysis.
- The system shall support controlled testing environments suitable for observational studies.

8.4 Software Requirements

The following software tools and platforms are required:

- Unreal Engine for developing the PC-based multiplayer game.
- Steam Online Subsystem for session management, networking, and voice communication.
- Flutter framework for developing the mobile companion application.
- Firebase Realtime Database for cloud-based data synchronization and event signaling.
- Android operating system for deploying the mobile application.
- Windows operating system for hosting and running the PC game.
- Version control system (e.g., Git) for collaborative development.

8.5 Hardware Requirements

The system requires the following hardware components:

- A Windows-based PC capable of running Unreal Engine applications.
- Multiple PCs or laptops for multiplayer client testing.
- Android smartphones or tablets for running the mobile companion application.
- Headphones or headsets with microphones for voice communication.
- Stable internet connectivity for real-time multiplayer and cloud communication.

9. Outcomes

The proposed research is expected to produce both technical and research-oriented outcomes that contribute to the understanding and application of game-based learning interventions for children with dyslexia.

9.1 Functional System Outcomes

- A fully functional multiplayer, voice-enabled literacy game integrated with a companion mobile application for letter tracing and interactive tasks.
- Successful implementation of real-time synchronization between the game environment and the mobile application using a secure, low-latency communication mechanism.
- A modular system architecture that supports multimodal interaction, including visual, auditory, and kinesthetic learning elements.
- Integration of an anticipatory feedback mechanism that provides immediate, non-punitive guidance during gameplay to normalize mistakes and support learning continuity.
- A reward-based progression system designed to encourage engagement and persistence without relying on competitive leaderboards.

9.2 Learning and Behavioral Outcomes

- Improved engagement and motivation among children with dyslexia during literacy-related activities through immersive gameplay.
- Increased willingness to participate socially in a virtual learning environment, supported by voice communication and cooperative gameplay mechanics.
- Reduced fear of failure as errors are framed as part of the learning process rather than as penalties.
- Strengthened attention and working memory through task sequencing, collaborative problem-solving, and sustained gameplay sessions.
- Reinforcement of letter recognition and ordering skills through repeated interaction with letters in contextualized word formation tasks.

9.3 Research and Evaluation Outcomes

- Empirical evidence comparing multimodal multiplayer gameplay with traditional single-player learning approaches for dyslexic children.
- Insights into how voice-enabled collaboration influences social confidence and communication behavior in a low-pressure virtual environment.
- Analysis of the effectiveness of reward-based motivation systems in maintaining engagement without increasing performance anxiety.
- Identification of usability challenges and design considerations specific to two-device interactive learning systems (PC + mobile).

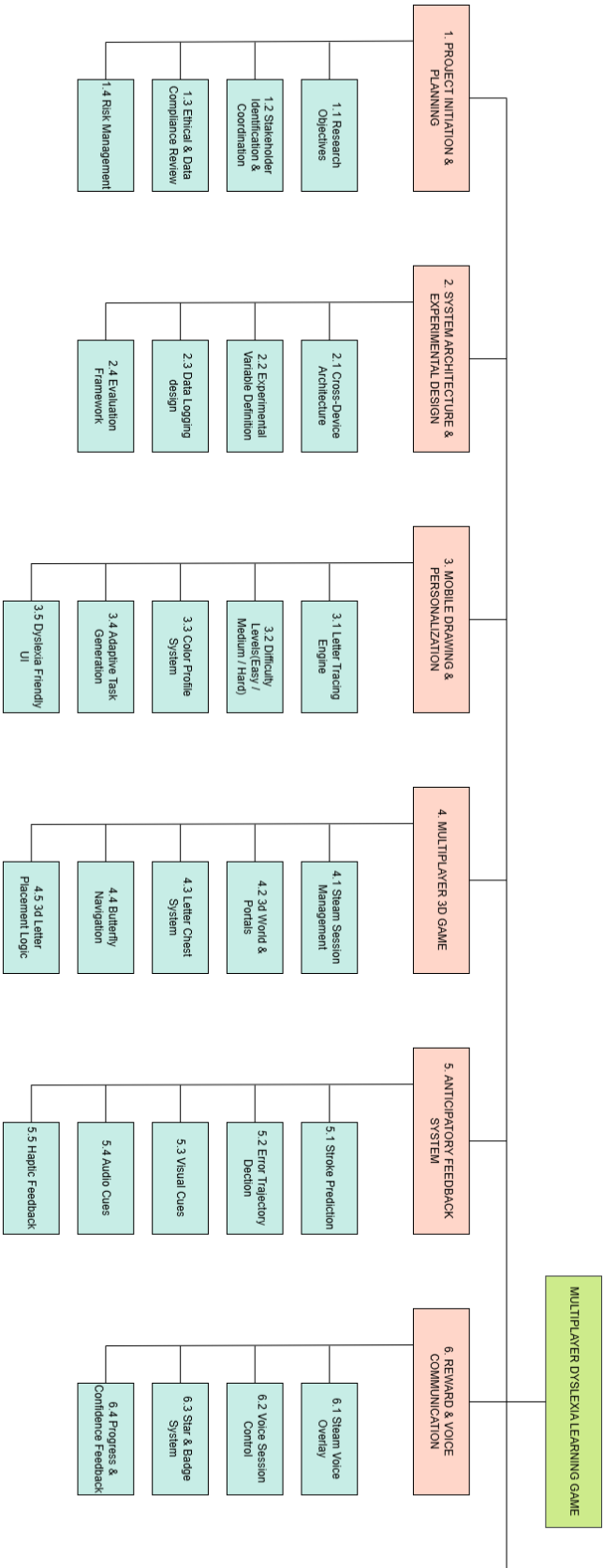
9.4 Academic and Practical Contributions

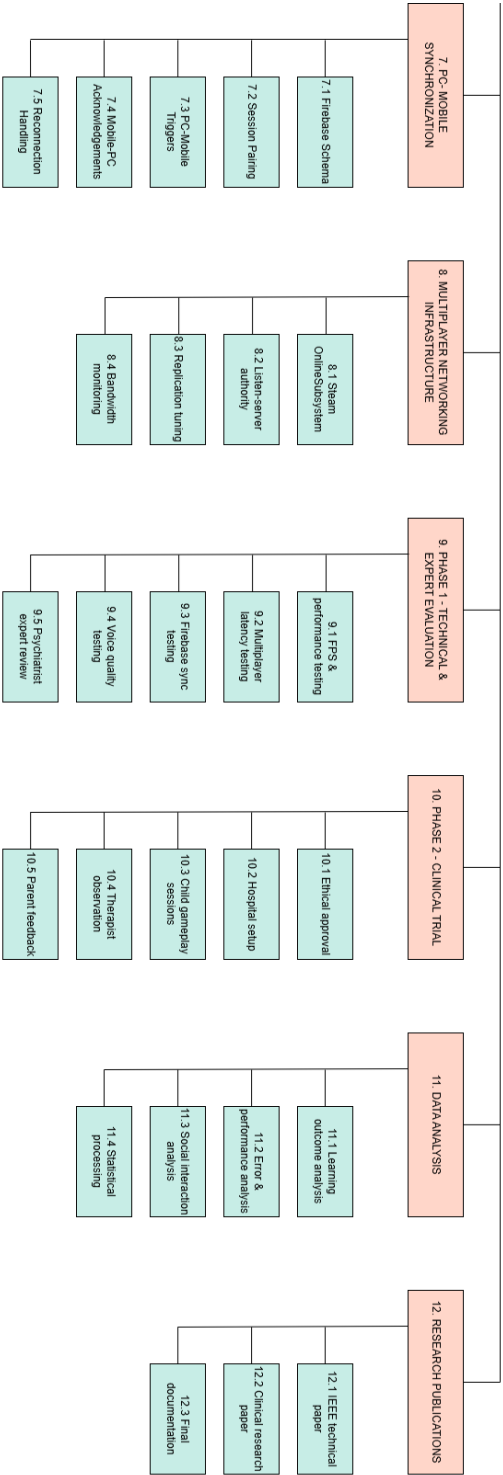
- A validated research prototype demonstrating the feasibility of a two-device, multiplayer learning game for dyslexia support.

- Design guidelines for developing dyslexia-sensitive multiplayer educational games that address both cognitive and socio-emotional needs.
- A foundation for future research involving clinical or educational trials in controlled settings.
- Potential applicability of the system framework to other learning difficulties or therapeutic learning contexts.

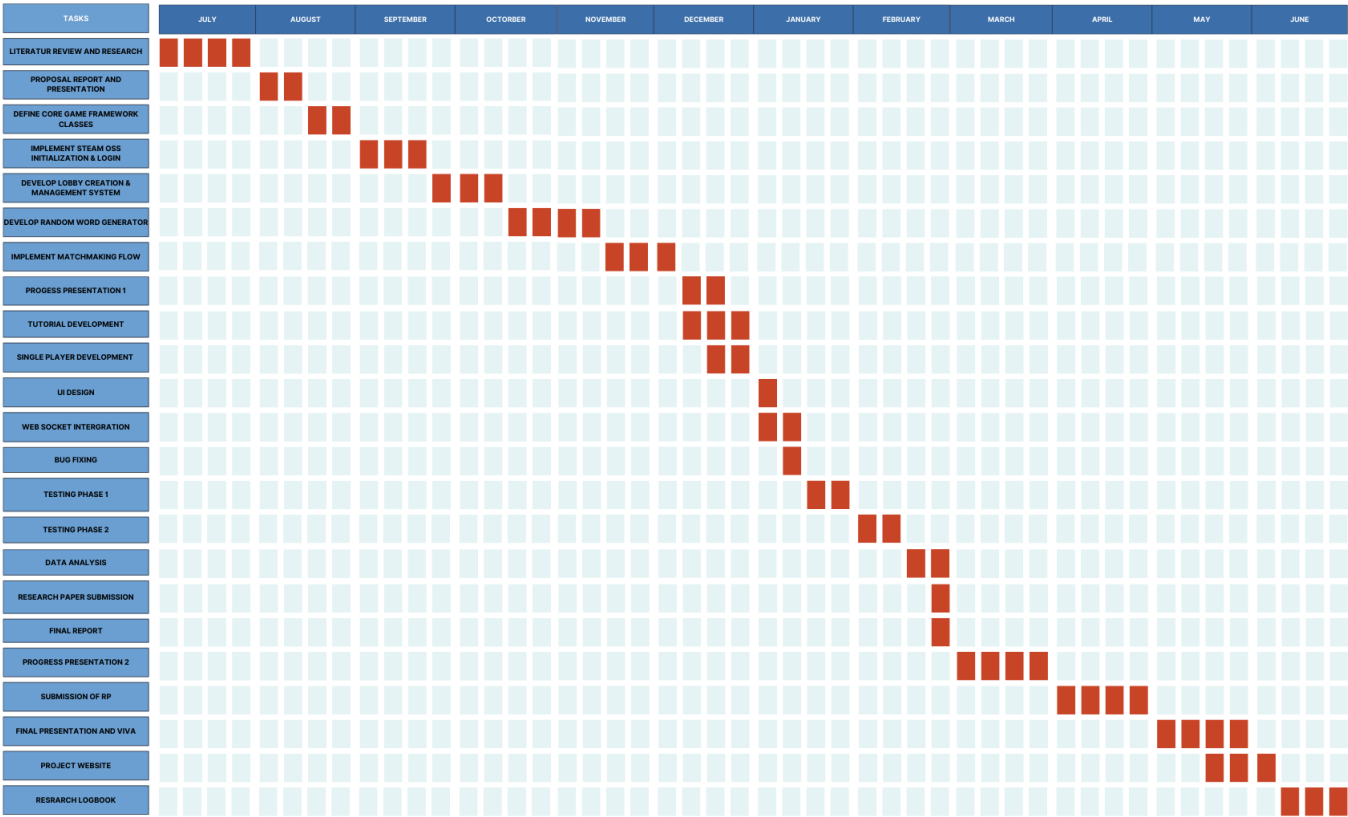
10. Project Commencement

10.1 Work Breakdown Structure





10.2 Gantt Chart



11. Commercialization

The commercialization strategy focuses on positioning the proposed multiplayer game as a supportive digital learning tool designed for children with dyslexia, suitable for both educational and supervised home environments.

- **Market Potential:**

With the increasing adoption of game-based learning and a growing emphasis on inclusive and assistive educational technologies in Sri Lanka and globally, there is a rising demand for interactive solutions that support children with learning differences. Digital tools that combine engagement with evidence-informed learning principles are particularly well aligned with current educational trends.

- **Target Audience:**

The primary target audience includes schools, special education units, therapy and learning support centers, and child development clinics. In addition, the system may be adopted by parents seeking structured, engaging, and accessible learning tools that can complement formal educational or therapeutic interventions at home.

- **Revenue Model:**

Potential revenue streams include institutional licensing models for schools and learning centers, subscription-based access for families, and partnerships with non-governmental organizations or government-led initiatives focused on learning disabilities and inclusive education. Further opportunities may arise through optional premium features such as extended content packs, customization options, and deployment on additional platforms.

- **Distinct Value Proposition:**

Unlike conventional educational games, the proposed solution integrates real-time multiplayer interaction, voice-enabled collaboration, and dyslexia-sensitive design principles within an immersive environment. By addressing not only literacy-related skills but also social interaction, motivation, and emotional confidence, the system offers a differentiated and holistic learning experience.

- **User Retention and Sustainability:**

Sustained engagement is supported through adaptive task difficulty, meaningful reward progression, and periodic content updates. These features are designed to maintain learner motivation while ensuring the system remains educationally relevant, enjoyable, and supportive over extended use.

12. Description of Personal and Facilities

- Facilitators: -
 - Mr. Didula Thanaweera Arachchi - Sri Lanka Institute of Information Technology
 - Mr. Nushkan Nisme - Sri Lanka Institute of Information Technology
 - Dr. Kasun Karunanayaka - University of Colombo School of Computing
 - Dr. Swarna Wijetunga - Lady Ridgeway Hospital for Children (LRH)
 - Dr. Darshani Hettiarachchi - Lady Ridgeway Hospital for Children (LRH)
- Facilities: -
 - Sirimavo Bandaranaike Specialized Children's Hospital, Kandy, Sri Lanka.

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